

Enhanced synthetic aperture radar image autofocus and classification using 2D SARNet framework

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ABSTRACT. A synthetic aperture radar (SAR) system is a notable source of information, recognized for its capability to operate day and night and in all weather conditions, making it essential for various applications. SAR image formation is a pivotal step in radar imaging, essential for transforming complex raw radar data into interpretable and utilizable imagery. Nowadays, advancements in SAR sensor design, resulting in very wide swaths, generate a massive volume of data, necessitating extensive processing. Traditional methods of SAR image formation often involve resource-intensive and time-consuming postprocessing. There is a vital need to automate this process in near-real-time, enabling fast responses for various applications, including image classification and object detection. We present an SAR processing pipeline comprising a complex 2D autofocus SARNet, followed by a CNN-based classification model. The complex 2D autofocus SARNet is employed for image formation, utilizing an encoder–decoder architecture, such as U-Net and a modified version of ResU-Net. Meanwhile, the image classification task is accomplished using a CNN-based classification model. This framework allows us to obtain near real-time results, specifically for quick image viewing and scene classification. Several experiments were conducted using real-SAR raw data collected by the European remote sensing satellite to validate the proposed pipeline. The performance evaluation of the processing pipeline is conducted through visual assessment as well as quantitative assessment using standard metrics, such as the structural similarity index and the peak-signal-to-noise ratio. The experimental results demonstrate the processing pipeline's robustness, efficiency, reliability, and responsiveness in providing an integrated neural network-based SAR processing pipeline.

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1 Introduction

Synthetic aperture radar (SAR) technology has become an important part of Earth's sensing systems allowing us to capture high-resolution images of the Earth's surface, under different conditions.¹ SAR data have proven invaluable across applications, such as monitoring the environment, responding to disasters, and conducting defense surveillance. SAR image formation is

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a pivotal step in radar imaging, essential for transforming complex raw radar data into interpretable and utilizable imagery.² This process is vital due to the complexity of SAR data, which encompasses both phase and amplitude information. The image formation in SAR employs specialized algorithms tailored to the synthetic aperture concept.³ These algorithms are designed to compensate for range variations caused by the movement of the radar platform, process Doppler shifts to enhance spatial resolution, preserve phase information for applications like interferometry, correct geometric distortions inherent in side-looking radar systems, and mitigate speckle noise.⁴

Over the years, there have been various techniques developed for SAR autofocus, such as the range Doppler algorithm (RDA) and chirp scaling algorithm.^{5,6} RDA is the most commonly used SAR image formation algorithm due to its simplicity, motion compensation, and flexibility.⁷ RDA enhances image formation and quality by employing pulse compression techniques in both range and azimuth directions, which improves the detection range and resolution of long pulses. This is achieved by modulating the transmitted pulse and correlating the received signal with it. The choice of algorithm depends on the specific application requirements.³

Due to the rapid advancements in deep learning and its application across diverse disciplines,^{8,9} the integration of deep-learning techniques in SAR image processing has emerged as a critical area of research.¹⁰ Exploration of machine learning methodologies, notably neural networks, for SAR autofocus and image refinement applications, has become increasingly prevalent. A research paper authored by Huo et al.¹¹ developed an SAR autofocus method leveraging a deep-learning-based generative adversarial network (GAN). This approach addresses SAR imaging defocus issues, enhancing focus effects compared to traditional methods. Utilizing deep learning, they built a complex GAN model and established training and test datasets with realistic SAR images. The results were objectively evaluated using image entropy, demonstrating the method's effectiveness and feasibility in SAR autofocus applications.

Rittenbach and Walters¹² represent an advancement, in the development of an SAR processing pipeline. The authors introduced a deep-learning framework known as RDAnet designed specifically for forming SAR images from raw radar data. By training RDAnet using echo data obtained from a mapping satellite, they successfully generated SAR images of quality, to those produced by conventional methods. According to the authors, this is the instance where real data has been used to demonstrate an integrated network-based SAR processing pipeline. Pu et al.¹³ introduced a deep-learning framework for SAR autofocus applications. Their research centers on integrating sparse autoencoder networks (SAE-Net) into SAR imaging, offering a novel solution for motion error correction and image optimization. This approach is significant as it handles SAR autofocus through a deep neural network. The effectiveness of this algorithm has been demonstrated through experiments conducted on both simulated and real-SAR data. The exploration of deep learning (DL) applications in SAR image formation remains an active area of research, presenting opportunities for increasing the ability to process SAR data rapidly from both spaceborne and airborne platforms for timely decision-making.¹⁴

1.1 Problem Definition

SAR image formation is a complex process that involves several steps, from acquiring raw data to creating a coherent two-dimensional image of the scene area.¹⁵ This process includes critical stages, such as range and azimuth compression, coupled with motion compensation to adjust for the movement of the platform.¹ The rapid development of SAR sensors that can cover very wide swaths generates a massive volume of data requiring extensive processing.¹⁶ This can lead to a delay in extracting meaningful information from SAR data, especially for time-sensitive applications that need a fast response like disaster management and security surveillance. To overcome this challenge, there is a focused effort on developing advanced techniques that enhance the speed and accuracy of SAR data processing, enabling near-real-time operations.¹² This includes the integration of deep learning with SAR signal processing, leading to cutting-edge solutions that outperform conventional methods in terms of time consumption and processing complexity. These advancements are crucial for rapid response applications and for fully leveraging the capabilities of SAR technology.

In this study, an SAR processing pipeline is introduced, which involves creating a valid dataset utilized to train an encoder-decoder network on the SAR image formation task.

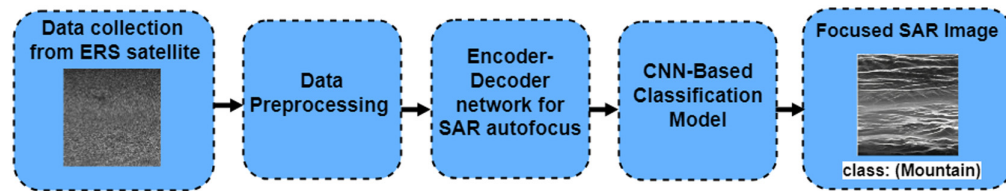


Fig. 1 SAR processing pipeline.

The results are used as a precursor to the CNN-based classification model, as illustrated in Fig. 1. This integration paves the way to get an end-to-end solution for SAR data processing. By incorporating DL into SAR processing pipelines, the needed near-real-time capabilities for applications demanding immediate responses are achieved. This paper presents its primary contributions to addressing these challenges as follows.

- Constructing a robust and efficient dataset from real data acquired from the European remote sensing (ERS) satellite and configuring this dataset for optimal use in both training and validation stages within the research framework.
- Developing a novel 2D-SARNet pipeline based on an encoder–decoder architecture for SAR image formation, followed by a CNN-based classification model.
- Conducting a comparative analysis with the state-of-the-art deep-learning models applied to SAR image processing, underscoring the enhanced efficacy of our proposed model.

The paper is structured as follows: Sec. 2 discusses the SAR image formation overview; Sec. 3 presents our methodology, which describes our network architecture; and Sec. 4 shows cases experimental results and provides a performance evaluation. Finally, Sec. 5 presents the conclusion.

2 SAR Image Formation Overview

The RDA is a crucial component in the SAR image formation process.⁵ It processes received raw data from a satellite to generate a focused and high-resolution SAR image. RDA was presented in 1978 as an algorithm that efficiently utilizes fast Fourier transforms (FFT) to condense the azimuth and range direction of the unfocused raw SAR data.¹ RDA consists of three main steps; range compression, range cell migration correction (RCMC), and azimuth compression as seen in Fig. 2. Each of these steps will be delved into in this section.⁷

In the outset, employing a linear FM chirp signal, SAR generates pulses to achieve high-resolution imaging through coherent transmission of wide-bandwidth signals.¹ The transformation of acquired echoes into intermediate frequency signals and their accumulation along the platform's flight path are integral steps in SAR's process, forming an extended antenna and enabling detailed imaging during motion. Equation (1) provides the definition of this signal:²

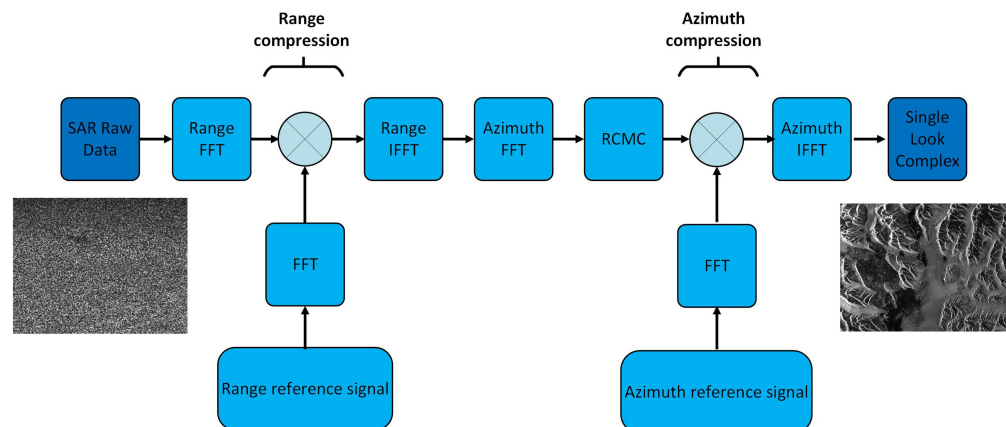


Fig. 2 Basic RDA block diagram.

$$s_{\text{pul}}(t) = \text{rect}\left(\frac{t}{T_p}\right) \cos(2\pi f_c t + \pi K_r t^2), \quad (1)$$

where f_c is the carrier frequency, t is the range time, and K_r is the chirp rate. With T_p serving as the pulse duration, the rectangular function $\text{rect}(\frac{t}{T_p})$ can be used to approximate the pulse envelope. The convolution of the pulse and the ground reflectivity g_{ref} , in the illuminated area, results in the received signal, or echo:

$$s_{\text{rec}}(t) = g_{\text{ref}}(t) * s_{\text{pul}}(t). \quad (2)$$

The received signal is determined by taking into account the time delay between the pulse's emission and reception, which is provided by $\frac{2R_b}{c}$, where R_b is the distance between the antenna and the target and M'_0 is the magnitude, the received signal is given by

$$s_{\text{rec}}(t) = M'_0 \text{rect}\left(t - \frac{2R_b}{c}\right) \cos\left(2\pi f_c \left(t - \frac{2R_b}{c}\right) + \pi K_r \left(t - \frac{2R_b}{c}\right)^2 + \Phi\right), \quad (3)$$

where Φ stands for a phase change that could be brought on by scattering caused by the reflecting surface's roughness.

The SAR system collects data by capturing echoes or radar signals that reflect from the Earth's surface.⁵ As the signal changes with azimuth time, it is now called $R_b(h)$. Before sampling, it is important to remove the radar carrier, which is represented as $\cos(2\pi f_0 t)$, through a demodulation process. This process produces the demodulated signal from a single point.² This received signal, known as $S_{\text{rec}}(t, h)$, provides information, about the reflectivity of the observed area and is typically sampled at azimuth positions over time:

$$\begin{aligned} S_{\text{rec}}(t, h) &= M_0 \text{rect}\left(t - \frac{2R_b(h)}{c}\right) w_{az}(h - h_c) \\ &\times \exp\left[-j4\pi \frac{f_c R_b(h)}{c}\right] \\ &\times \exp\left[j\pi K_r \left(t - \frac{2R_b(h)}{c}\right)^2\right]. \end{aligned} \quad (4)$$

The azimuth antenna pattern is represented by w_{az} , the azimuth time by h , and the complex constant M_0 is represented by $M_0 = M'_0 \exp(j\Phi)$.

As shown in Fig. 2, the RDA starts with range compression along the range direction followed by a fast convolution.¹⁵ This sequence encompasses a range FFT, multiplication with a matched filter in the frequency domain, and culminates in a range inverse fast Fourier transform (IFFT).⁴ Assuming $S_0(f_t, h)$ is the range FFT of s_r and $G(f_t)$ is the frequency domain matched filter, using the received demodulation signal provided by Eq. (4), the output of this stage is obtained by

$$S_{rc}(t, h) = \text{IFFT}_t[S_0(f_t, h)G(f_t)] = M_0 \rho_r \left(t - \frac{2R_b(h)}{c}\right) w_{az}(h - h_c) \exp\left[-j4\pi \frac{f_c R_b(h)}{c}\right], \quad (5)$$

where the IFFT of the rectangular function ρ_r is the compressed pulse envelope.

An azimuth FFT translates the data into the range-Doppler domain.¹ Since the first exponential of Eq. (5) is fixed for each target and $f_h = -K_a h$, where K_a is the azimuth FM rate of the point target signal, the output after the azimuth FFT is provided by

$$\begin{aligned}
S_{rd}(t, f_h) &= \text{FFT}_h[S_{rc}(t, h)] = M_0 \rho_r \left(t - \frac{2R_{rd}(f_h)}{c} \right) W_{az}(f_h - f_{hc}) \\
&\quad \times \exp \left[-j4\pi \frac{f_c R_b(h)}{c} \right] \\
&\quad \times \exp \left[j\pi \frac{f_h^2}{K_a} \right], \tag{6}
\end{aligned}$$

where W_{az} is the antenna's beam pattern's Doppler spectrum envelope.

The RCMC process is crucial for compensating for range variations in data due to platform movement, known as range migration.⁵ This involves adjusting the data in memory to correct the trajectory, facilitating the process of azimuth compression along each parallel azimuth line.² This correction straightens and corrects the data path. RCMC is a vital step in radar signal processing, ensuring accurate and coherent representation of spatial information despite platform motion.¹

$$\delta R(f_h) = \frac{\lambda^2 R_b f_h^2}{8V_r^2}, \tag{7}$$

where λ is the wavelength of carrier frequency f_c , resulting in the following signal:

$$S_1(t, f_h) = M_0 \rho_r \left(t - \frac{2R_b}{c} \right) W_{az}(f_h - f_{hc}) \exp \left[-j4\pi \frac{f_c R_b(h)}{c} \right] \exp \left[j\pi \frac{f_h^2}{K_a} \right]. \tag{8}$$

The aim of azimuth compression is to focus the trajectory's energy into a cell along the azimuth direction.² Following RCMC, a matched filter is applied to the data, represented in the frequency domain. The subsequent step involves performing an IFFT. The matched filter in the frequency domain can be represented as

$$H_a(f_h) = \exp \left[-j\pi \frac{f_h^2}{K_a} \right]. \tag{9}$$

The resultant signal from azimuth compression is given by

$$S_2(t, f_h) = S_1(t, f_h) H_a(f_h) = M_0 \rho_r \left(t - \frac{2R_b}{c} \right) W_{az}(f_h - f_{hc}) \exp \left[-j4\pi \frac{f_c R_b}{c} \right], \tag{10}$$

where ρ_a is the amplitude of the azimuth impulse response.

Finally, we apply an azimuth IFFT to convert the data into the time domain resulting in a compressed image. After this step, we represent the compressed image as

$$S_{az}(t, h) = \text{IFFT}_h[S_2(t, f_h)] = M_0 \rho_r \left(t - \frac{2R_b}{c} \right) \rho_a(h) \exp \left[-j4\pi \frac{f_c R_b}{c} \right] \exp[j2\pi f_{hc} h]. \tag{11}$$

To summarize, the RDA plays a significant role in generating SAR images, effectively converting raw radar data into SAR images serving as a foundation for analysis and processing.² In Sec. 3, SAR images acquired from the RDA will be utilized as true information for our dataset. Using the insights extracted from these images, an approach will be developed that employs the encoder–decoder network for SAR autofocus refinement. Our primary objective is to integrate the SAR autofocus model with a classification model, thereby establishing a comprehensive SAR processing pipeline for near-real-time applications across various fields. This integrated pipeline, 2D-SARNet, is designed to enhance the efficiency and speed of SAR data autofocus, making it suitable for real-world scenarios, such as disaster management and defense applications. Furthermore, the development of 2D-SARNet, trained to replicate the entire RDA process, addresses the challenge of SAR image formation as a supervised learning task. This demonstrates the advanced integration of deep-learning methodologies in refining and analyzing SAR imagery.

3 Methodology

This section outlines the framework and techniques used to address our research objectives, encompassing data collection and preprocessing, the SAR autofocus model, and the CNN-based classification model, as depicted in Fig. 3. The proposed pipeline has been tested and validated,

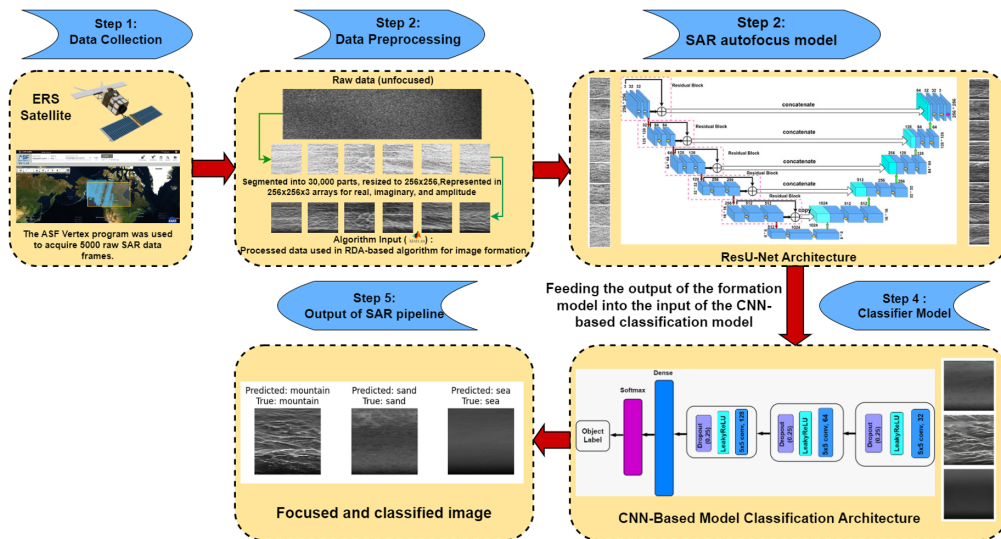


Fig. 3 Proposed SAR pipeline framework process.

with results compared to the state-of-the-art models from previous studies, such as convolutional autoencoders (CAEs),¹² sparse autoencoders (SAEs),¹³ and GAN networks,¹¹ as discussed in the related work. These comparisons utilized the same constructed datasets to assess the suitability and performance of the pipeline in SAR image processing tasks.

The U-Net architecture, originally designed for image segmentation,¹⁷ has been adapted for the SAR autofocus task by adding depth with extra encoder-decoder layers, adjusting the output layer to a three-channel configuration with a sigmoid activation function, and using mean-squared-error (MSE) loss function for model compilation.¹⁸ These adaptations enhance its capability for detailed image reconstruction, shifting its application from segmentation to SAR autofocus tasks.¹⁷ A more sophisticated U-Net architecture has been achieved by integrating a residual block within its encoder path, creating the ResU-Net. This modification allows for deeper feature extraction, resulting in higher accuracy.

3.1 Data Collection and Preprocessing

In this study, raw SAR echo data collected by the ERS satellite,¹⁹ which was operational from 1995 to 2011, were utilized. These data are accessible through the Alaska Satellite Facility (ASF),²⁰ which provides academic researchers with both raw and processed remote sensing data from a constellation of satellites, including SAR data. To obtain raw SAR data frames for a specific area of interest (AOI), the ASF Vertex program was employed,²¹ a process that is depicted in Fig. 4. This tool is specifically designed for downloading SAR data frames from

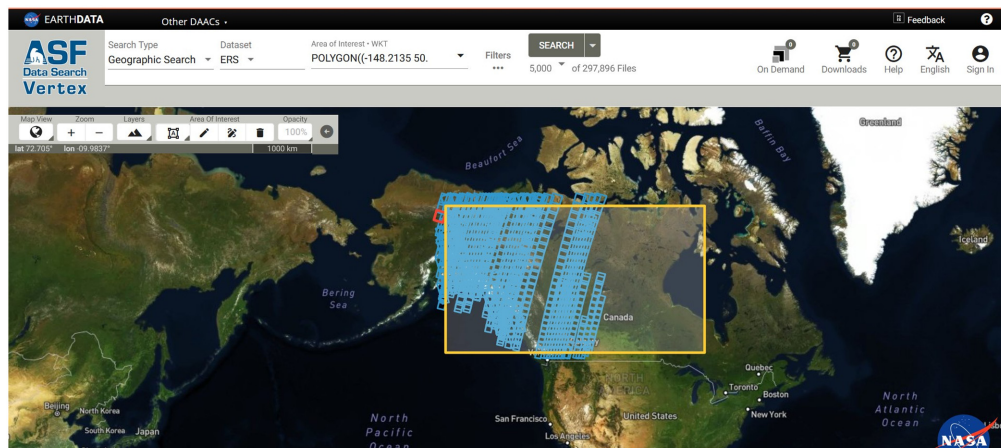


Fig. 4 ASF vertex tool for the AOI.

Algorithm 1 Data preprocessing for SAR autofocus pipeline.

```

1: for each raw_file in Raw_files_list do
2:   Extract parameters from corresponding LDR file:
3:   fs,fc,prf,τ,bw,v,ro,fd ← ExtractParameters(ldr_files_list[raw_file])
4: end for
5: for each raw_file in Raw_files_list do
6:   Load raw SAR data:
7:   rawData ← LoadRawData(raw_file)
8:   rows,cols ← GetDimensions(rawData)
9:   tileSize ← 4096
10:  numTilesRows ←  $\left\lfloor \frac{rows}{tileSize} \right\rfloor$ 
11:  numTilesCols ←  $\left\lfloor \frac{cols}{tileSize} \right\rfloor$ 
12:  for r in 1 to numTilesRows do
13:    for c in 1 to numTilesCols do
14:      tile ← ExtractTile(rawData, r, c, tileSize)
15:      normalizedTile ← Normalize(tile)
16:      resizedInput ← Resize(normalizedTile, 256×256)
17:      focusedImage ← ApplyAutofocus(normalizedTile, fs, fc, prf, τ, v, ro, fd)
18:      resizedOutput ← Resize(focusedImage, 256×256)
19:      SaveImage(resizedInput, outputDirectory, "input", raw_file, r, c)
20:      SaveImage(resizedOutput, outputDirectory, "output", raw_file, r, c)
21:    end for
22:  end for
23: end for

```

the ERS satellite archives. The ERS collected mapping data; therefore, SAR images used in this work consist of diverse landscapes, including mountains, sea, and sand. This dataset serves as the foundation for our study, allowing for a comprehensive analysis of SAR autofocus and classification task.

The dataset comprises 5000 ERS raw images with I and Q channels, capturing diverse landscapes. These feed into the RDA to focus the raw SAR echo data as mentioned in Sec. 2, which resulted in generating output datasets.²² This extensive preprocessing extracted useful information from the raw SAR data to train SAR autofocus models and a CNN-based classification model, enabling the network to effectively learn how to autofocus and classify. As part of this study, a defined algorithm was created to handle the data preparation and organization tasks. The following pseudocode encapsulates the key components of Algorithm 1, ensuring systematic handling of the data for subsequent stages in the SAR autofocus pipeline.

3.2 Network Architecture

In this section, the network structure that powers the SAR pipeline is explored, providing insights into the details of its components:

3.2.1 SAR autofocus model: U-Net architecture

U-Net architecture, initially developed for biomedical image segmentation, has been effectively tailored for SAR Autofocus applications,²³ addressing the unique challenges of SAR imagery by

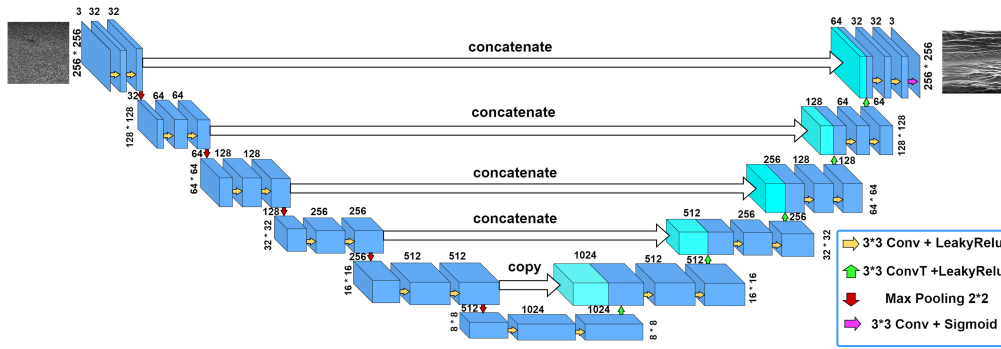


Fig. 5 U-Net architecture.

leveraging its encoder–decoder structure with skip connections as depicted in Fig. 5. This design is integral for capturing and preserving spatial information across various resolutions, which is vital for detailed feature extraction and image reconstruction in our SAR autofocus model. The encoder uses convolutional layers with leaky ReLU activation and max pooling to robustly capture complex features and reduce dimensionality, crucial for autofocus correction. The decoder, featuring transposed convolutions and integrating skip connections, gradually reconstructs image resolution, enhancing spatial detail. Modifications, such as a sigmoid activation in the final layer, optimize the architecture for SAR, significantly improving image clarity and autofocus accuracy by correcting inherent phase errors in SAR images.

In the U-Net SAR autofocus model, the contracting path, also known as the encoder, plays a role in extracting features, from the input SAR image.²⁴ It consists of repeated blocks containing two convolutional layers followed by a max-pooling layer. The convolutional layers, using 3×3 filters with leaky ReLU activation and “same” padding, progressively increase the number of filters from 32 to 512, allowing the network to extract increasingly complex features from the input. The max-pooling layers, with a 2×2 window, reduce the spatial dimensions after each block, leading to a compressed and abstracted representation of the input.²⁵

Let E_i represent the feature map at the i 'th level within the encoder, for $i = 1, 2, \dots, 5$. Then, each E_i can be described as

$$E_i = \begin{cases} \text{leakyReLU}(\text{Conv}_i(X)), & \text{for } i = 1 \\ \text{leakyReLU}(\text{Conv}_i(\text{MP}(E_{i-1}))), & \text{for } i > 1 \end{cases} \quad (12)$$

Here Conv_i denotes the convolutional layer specific to level i . The leaky ReLU activation is defined as follows:

$$f(x) = \begin{cases} x, & \text{if } x > 0 \\ \alpha x, & \text{if } x \leq 0 \end{cases} \quad (13)$$

In this equation, the variable x represents the input to the activation function while α is a constant that is usually assigned a value such as (0.1). The leaky ReLU ensures that negative inputs have a gradient thereby avoiding the issue known as “dying ReLU.”²⁵

The U-Net SAR autofocus model bottleneck features two 3×3 convolutional layers with 1024 filters each, using leaky ReLU activation and same padding. The bottleneck, denoted as M , focuses on extracting features at a specific level through two layers:

$$M = \text{leakyReLU}(\text{Conv}_{i+1}(\text{Conv}_{i+2}(E_i))). \quad (14)$$

In the U-Net SAR autofocus model, the expanding path, also known as the decoder uses transposed convolutions (denoted as Deconv) with decreasing filter sizes from 512 to 32 to upscale feature maps.¹⁷ Each of these layers is followed by concatenation with the corresponding encoder output, adding back lost spatial information. This is then refined by two standard convolutional layers with leaky ReLU. Finally, a transposed convolution with three filters reconstructs the image to its original resolution. Let D_i represent the feature map at the i 'th level within the decoder, for $i = 1, 2, \dots, 5$. Then each D_i can be described as

$$D_i = \begin{cases} \text{leakyReLU}(\text{Deconv}_i(M)), & \text{for } i = 1 \\ \text{leakyReLU}(\text{Deconv}_i(\text{Concat}(\text{LeakyReLU}(D_{i-1}), E_{k-i}))), & \text{for } i > 1 \end{cases} \quad (15)$$

In this context, Deconv_i is the deconvolution operation at level i . The Concat operation combines a feature map with its feature map from the encoder. E_{k-i} is the feature map from the encoder at the corresponding level.

The final layer of the U-Net, originally designed for segmentation tasks, requires modification to adapt it for the SAR image autofocus application. This layer employs a sigmoid activation function to transform the decoder's output into the SAR image. This conversion involves applying the sigmoid activation function to the result obtained from the layer.²⁶ Using a sigmoid function in image formation tasks ensures pixel values stay within a valid range, enhancing image quality and realism. It also stabilizes training by avoiding unrealistic pixel values. This layer is composed of a convolutional layer followed by the sigmoid activation function, which together produce the SAR image. The sigmoid activation function is defined as follows:²⁷

$$f(x) = \frac{1}{1 + e^{-x}}. \quad (16)$$

In this equation, let us use x to represent the input, to the sigmoid function while e will stand for the base of the logarithm, it is also a mathematical constant.

In conclusion, the U-Net architecture has been effectively adapted for SAR autofocus through key modifications, including the addition of an extra encoder and decoder layer to increase network depth, and the adjustment of the output layer to a three-channel configuration with a sigmoid activation function, signifying a shift toward direct SAR image reconstruction. The choice of an MSE loss function for model compilation supports the precise reconstruction of SAR images. These alterations collectively shift the focus of the U-Net architecture from its traditional application in segmentation to meet the specific demands of SAR autofocus, thereby enhancing its ability to correct image distortions and improve the clarity and resolution of SAR imagery.

3.2.2 SAR autofocus model: ResU-Net architecture

The ResU-Net architecture enhances the foundational U-Net model by incorporating residual blocks (ResBlocks) into the encoder path,²⁸ significantly boosting its ability to learn and retain key features from SAR images, crucial for autofocus applications where precise phase correction and image clarity are essential. These ResBlocks feature “shortcut” connections that bypass layers, promoting direct gradient propagation and effectively combating the vanishing gradient problem, thus improving training efficiency as depicted in Fig. 6. This modification is especially valuable in deep-learning models for SAR autofocus, as it allows the network to utilize detailed spatial information from deeper layers, dramatically enhancing autofocus performance.

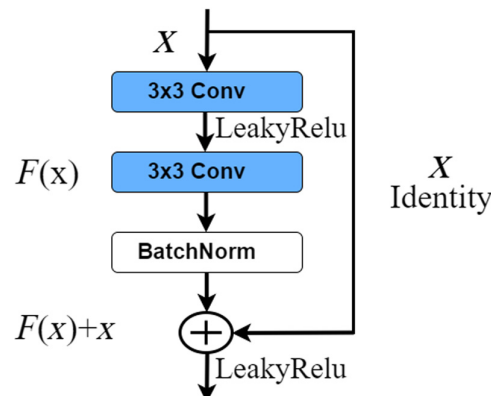


Fig. 6 Residual learning: a building block.

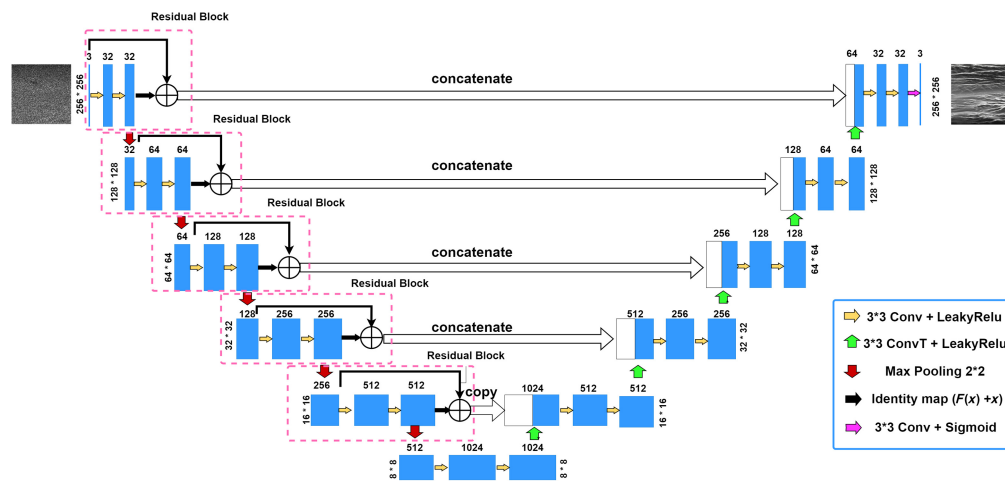


Fig. 7 Modified U-Net with residual blocks.

By embedding ResBlocks within the encoder, the ResU-Net architecture not only maintains robust feature representation across all network levels but also optimizes phase error correction and image quality, leading to superior autofocus outcomes.²⁴

In this SAR Autofocus model, with residual blocks, connections are introduced that help in extracting features and enable smooth gradient flow during the training process.²⁹ This modification is specifically implemented in the encoder path, where each is shown in Fig. 7. This is particularly valuable for complex tasks like SAR autofocus, enhancing feature learning efficiency, and aiding convergence in time-sensitive applications.

The modified U-Net design, which includes residual blocks maintains a structure to the traditional U-Net.²³ However, it incorporates these blocks, within the encoder pathway. In this approach, multiple residual blocks are used in the encoder to extract features from the SAR image.²⁹ This ensures that important features are preserved. As the encoder progresses it systematically reduces dimensions while retaining information. The U-Net SAR Autofocus model improves its encoder path by replacing each layer with a block resulting in enhanced feature extraction.²⁸ This modified architecture effectively strengthens feature learning in the encoder while maintaining the core structure of U-Net. These improvements collectively enhance the model's ability to detect and phase errors, in SAR images.

Similar to the traditional U-Net, skip connections establish connections between corresponding encoder and decoder layers to facilitate the transfer of fine-grained information.²³ The skip connections are created by concatenating the encoder output with the corresponding decoder output. The decoder consists of transposed convolutional layers (deconvolutional layers) to upsample the feature maps.¹⁷ The equations for the decoder layers remain the same as those in the traditional U-Net. The final output layer applies a sigmoid activation function to the decoder's output to generate the focused SAR image.

3.2.3 CNN-based model classification architecture

The CNN-based classification model is used for SAR image scene classification by employing convolutional layers with 5×5 kernels and strides to effectively extract spatial features necessary for categorizing different terrains.³⁰ To increase robustness and prevent overfitting, it incorporates leaky ReLU activations and dropout regularization within each layer. The features are then flattened and passed through fully connected layers, ending in a SoftMax function for precise terrain classification, thus improving the interpretability and utility of SAR data.

The classifier network is composed of three convolutional layers, each utilizing 5×5 convolutional kernels and a stride of two in each dimension. The first layer has 32 filters, the second has 64 filters, and the third has 128 filters. All layers are activated using the leaky ReLU activation function and dropout layers for regularization as depicted in Fig. 8.

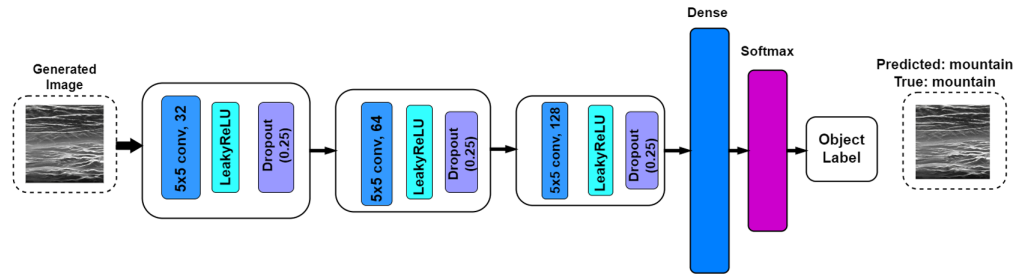


Fig. 8 Classification model architecture.

Each convolutional layer can be represented as follows:

$$\begin{aligned} \text{convolutional layer } i: \text{output}_i &= \text{Conv}_i(\text{input}_i) \\ \text{output}_i &= \text{leaky ReLU}(\text{output}_i, \alpha = 0.2) \\ \text{output}_i &= \text{dropout}(\text{output}_i, p = 0.25). \end{aligned} \quad (17)$$

In this context, for $(i = 1, 2, 3)$, Conv_i denotes the convolutional layer at position i . The term α refers to the slope parameter of the leaky ReLU activation function, whereas p signifies the probability of applying dropout. Initially, for the first layer, the input (input_1) is set as X . For layers beyond the first ($i > 1$), the input to each layer (input_i) is the output from the previous layer (output_{i-1}).

After the convolutional layers, the feature maps are flattened to create a one-dimensional vector, which serves as input to the fully connected layers.³¹ The flattening operation can be expressed as

$$\text{flattened features} = \text{flatten}(\text{output}_i). \quad (18)$$

The final classification decision is made by the fully connected layer, which includes a dense layer utilizing a SoftMax activation function to produce class probabilities. The fully connected layer can be expressed as follows:

$$\text{class probabilities} = \text{SoftMax}(\text{dense}(\text{flattened features})). \quad (19)$$

This SoftMax activation function denoted as “ σ ” assigns probabilities to each class,²⁶ allowing the model to determine its class predictions, which can be represented as follows:

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_k e^{z_k}} \text{ for each element } z_i \text{ in the vector } z. \quad (20)$$

3.2.4 Loss function and optimization

For training the ResU-Net model, the MSE loss function is commonly employed for regression tasks.¹¹ The MSE loss quantifies the discrepancy between the predicted image $\hat{Y}$ and the ground truth image Y as follows:

$$L(Y, \hat{Y}) = \frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2, \quad (21)$$

where N is the total number of pixels in the image. Y_i is the ground truth pixel value, and $\hat{Y}_i$ is the predicted value.

The optimization process aims to minimize this loss. The adaptive moment estimation (Adam) optimizer is a popular choice for optimizing model weights.³² It updates the weights using the following rule:

$$W_{t+1} = W_t - \frac{\alpha}{\sqrt{\hat{v}_t + \epsilon}} \cdot \hat{m}_t. \quad (22)$$

The Adam optimizer updates a model's weight W_t at iteration t by adjusting it based on the learning rate α , as well as bias-corrected estimates of the first and second moments of the gradients, denoted as $\hat{m}_t$ and $\hat{v}_t$, respectively. These moments are crucial for adapting the learning rate for each weight during optimization.³² Additionally, a small number ϵ is used to prevent division by zero.

The sparse categorical cross-entropy loss function is utilized in the classification model, a common choice for single-label classification tasks.¹⁸ This loss function quantifies the dissimilarity between the predicted class probabilities and the true class labels. It can be mathematically represented as follows:

$$L(y, \hat{y}) = - \sum_{i=1}^N y_i \log(\hat{y}_i), \quad (23)$$

where $L(y, \hat{y})$ represents the loss between the true label y and the predicted label $\hat{y}$, N denotes the total number of samples, y_i corresponds to the true label for the i 'th sample, represented as an integer, and $\hat{y}_i$ signifies the predicted probability for the class of the i 'th sample.

3.3 Model Training and Hyperparameters

The SAR processing pipeline uses two models: a complex 2D autofocus model (SARNet) based on U-Net/ResU-Net architectures and a CNN-based landscape classification model. The complex 2D-SARNet is used for autofocus SAR images, whereas the CNN model classifies terrains such as mountains, sand, and sea. These models are trained separately, not in a unified process. First, the ResU-Net for autofocus is trained, followed by the CNN for landscape classification. This sequential training ensures that each model is finely tuned for its specific task. The training procedures of the 2D-SARNet framework are detailed in a flowchart, demonstrating the two-stage process as depicted in Fig. 9.

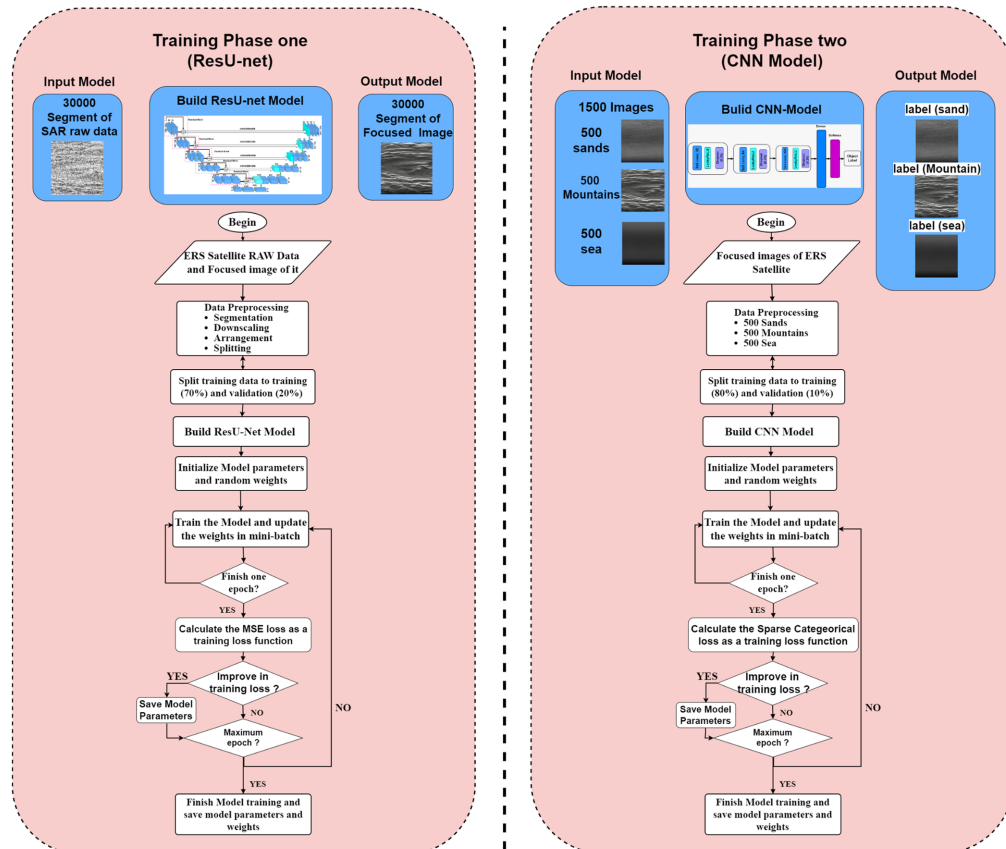


Fig. 9 2D-SARNet framework training flowchart.

Table 1 Hyperparameters of SAR processing pipeline.

Parameters	UNet/ResU-Net	CNN classification
Data split	Train/valid/test 70%/20%/10%	Train/valid/test 80%/10%/10%
Optimizer	Adam	Adam
Learning rate	0.0001	0.0001
Loss function	MSE	Sparse categorical
Epochs	100	20
Batch size	64	32
Early stopping	Yes	Yes
Data shuffled	Yes	Yes

The dataset employed to train the ResU-Net/U-Net models included 30,000 segments of SAR raw data along with 30,000 segments of focused images, which served as ground truth images. These focused images were obtained using the traditional RDA. For the classification task, the utilized dataset comprised a total of 1500 images, ensuring a balanced representation of different landscapes: 500 images of mountainous landscapes, 500 images showcasing sea landscapes, and 500 images depicting sandy landscapes. The training strategy and hyperparameters for each model are outlined in Table 1.

Figure 10 displays a sample of 64 minibatches, which represent samples from both the training input datasets and training output datasets for the ResU-Net/U-Net models.

In Fig. 11, the model loss trajectories of U-Net and ResU-Net architectures reveal a notable superiority of the ResU-Net. The ResU-Net indicates a more significant convergence. These results underscore the effectiveness of integrating residual connections into the U-Net framework. By doing so, the model reinforces its ability to preserve crucial features, leading to more accurate autofocus results in SAR imagery processing. The ResU-Net superior convergence indicates its heightened efficiency in learning complex patterns and highlights its potential for superior performance in comparison to its U-Net counterpart. This observation emphasizes the pivotal role of residual connections in refining model capabilities for advanced image processing tasks.

Similarly, the model trajectories for the CNN-based classification model are shown in Fig. 12. The accuracy curve for both training and validation datasets indicates fast training and

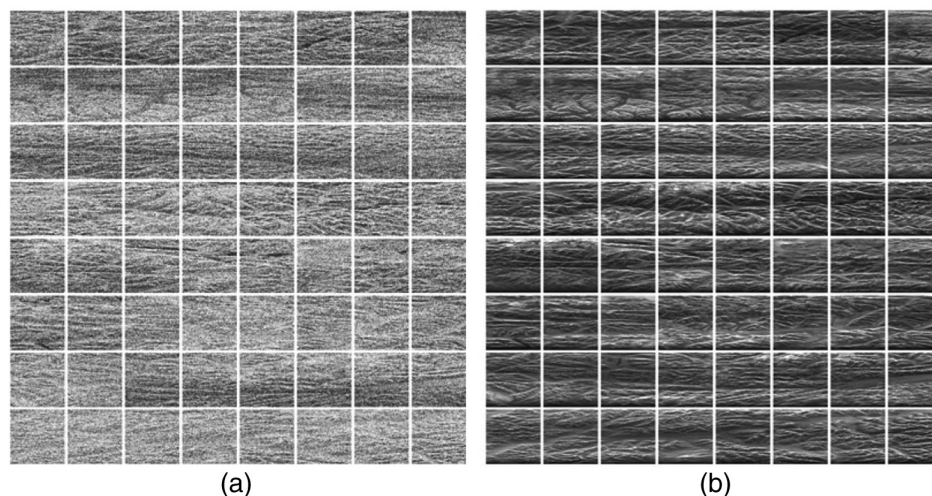


Fig. 10 Example of a sample of 64 mini-batches in ResU-Net/U-Net models: (a) training input batch and (b) training output batch.

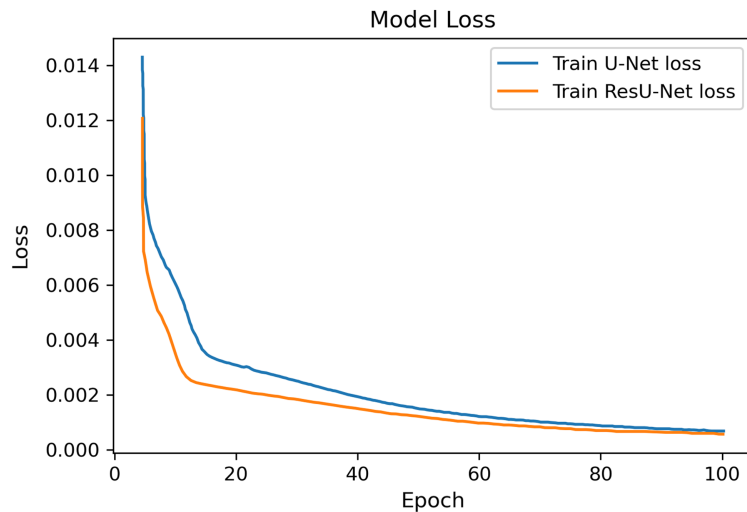


Fig. 11 Training loss across epochs for both the U-Net and ResU-Net models.

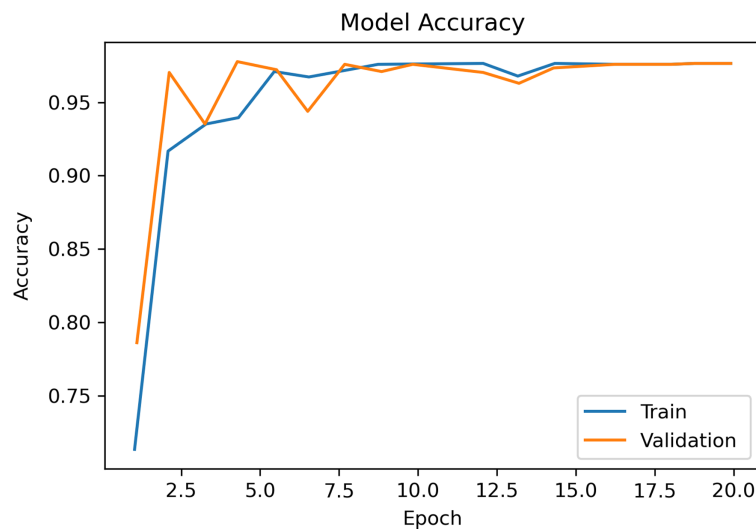


Fig. 12 Accuracy curve illustrating the training performance of the CNN-based model.

effective learning with minimal signs of overfitting. The results show successful training for landscape classification.

Overall, the integrated SAR pipeline combines a ResU-Net for image focusing with a CNN-based classification model for landscape categorization, creating a comprehensive SAR processing pipeline. The chosen hyperparameters and training strategies have been utilized to enhance the framework's effectiveness. This approach aims to ensure the pipeline's robustness and reliability, avoiding overfitting to ensure accurate and effective interpretation of intricate SAR information. Such a pipeline is designed to deliver consistent and stable results in critical situations where reliable performance is crucial in real-world SAR data processing scenarios.

4 Experimental Work and Performance Evaluation

4.1 Dataset and Experimental Setup

The dataset consists of 5000 ERS raw image data, each consisting of I and Q channels, accompanied with its metadata, downloaded from the ASF official website.²⁰ Figure 4 showing the downloaded window. The dataset covers a variety of landscapes, such as mountains, sand, and sea, each scene frame spans a 100 km swath with a spatial resolution of (≤ 30 m in range and

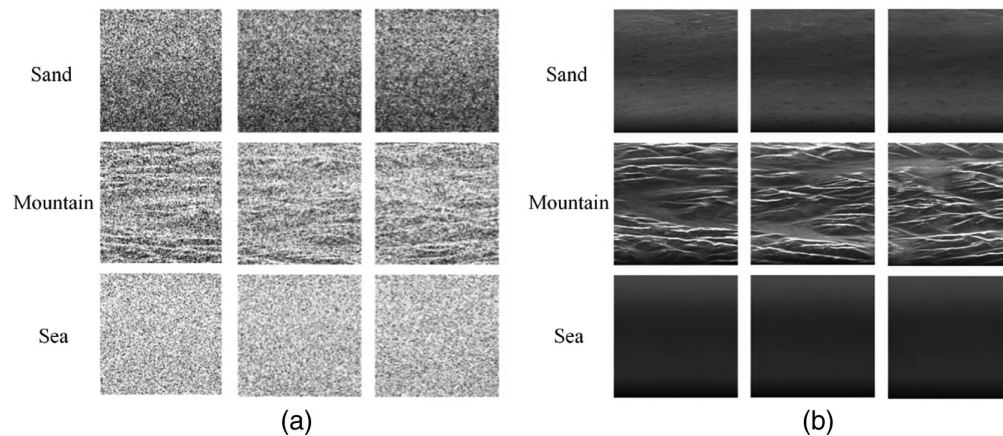


Fig. 13 Training dataset: (a) raw image data tiles and (b) focused image using RDA.

≤ 26.3 m in azimuth).¹⁹ To accommodate the processing resources, each scene frame is divided into 6 segments with a size (4096×4096 pixels) resulting in a total of 30,000 segments. The segments are then downsampled to 256×256 for efficient computing. The preprocessing stage includes data normalization, which is done by dividing each channel by its complex magnitude to confine values within $[-1, 1]$ and centering around zero by subtracting the channel mean from each pixel. The processed data were then represented as arrays of size $256 \times 256 \times 3$. Each array had three channels: one for the real part, another for the imaginary part, and a third for the amplitude. These data are passed to the RDA-based algorithm to perform the formation task as mentioned in Sec. 2. Figure 13 showcases preprocessed samples across different terrains. To optimize model training and evaluation, the 30,000 segments were split into training, validation, and testing datasets. This structured approach was crucial for developing an effective model capable of interpreting complex data.

4.2 Comparative Analysis of SAR Autofocus Models

This study provides an analysis that compares the proposed ResU-Net model with existing state-of-the-art models, such as CAE, SAE, and GANs. This comparison specifically focuses on evaluating the abilities of ResU-Net for SAR autofocus tasks. The evaluation thoroughly examines the performance of each model using the same dataset to ensure fairness and highlights the strengths of ResU-Net in comparison to these established models.

The RDAnet-based autoencoder network, as discussed in Ref. 12, uses a convolutional encoder with four blocks, filters increasing from 64 to 512, linked by leaky ReLUs and 2D max pooling (stride 2). Spatial dropout is added after the final block to prevent overfitting. It includes two fully connected layers with 2014 units each. A key feature is the integration of a deep residual network with 16 residual blocks for enhanced image resolution. The SAE-Net, as discussed in Ref. 13, includes an encoder using the alternating direction method of multipliers algorithm for efficient SAR data processing and a decoder that inverses the imaging process. Additionally, the SAE-Net for SAR autofocus incorporates two sparsity constraints: L1-norm regularization, which promotes sparse weight representation by minimizing less important weights, and image sharpness regularization, integrated into the loss function, to enhance the sharpness of the reconstructed image. As discussed in Ref. 11, the network architecture is a GAN with a complex generator and discriminator. The generator, which is modified based on the super-resolution generative adversarial networks model, comprises three convolutional layers. These layers maintain the same size for both input and output images and include a complex residual module to enhance SAR image processing. The discriminator is structured into three parts: convolution layers, batch normalization, and complex leaky ReLU activation functions.

In Fig. 14, the loss curves of various models are depicted, showcasing their performance in the autofocus of SAR imagery. Each model's curve provides insight into its ability to minimize loss related to the reconstruction error of SAR images. Among these models, the ResU-Net

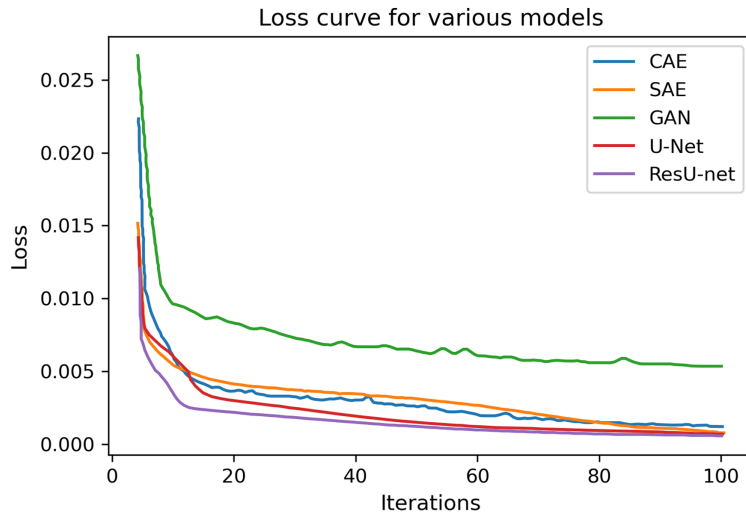


Fig. 14 Comparative loss curves across epochs during the training stage.

model shows a superior performance, with its loss curve compared to the others. This indicates that it converges effectively due to its architectural feature of utilizing residual connections. These connections play a role in preserving features and ultimately leading to more accurate SAR image formation.

On the other hand, the U-Net model also demonstrates a superior performance but is slightly lower than the enhanced structure of ResU-Net, which effectively captures complex patterns. CAE is competitive but does not surpass the performance of the U-Net derivatives. Meanwhile, the SAE registers a consistently higher loss throughout the iterations. Finally, the GANs exhibit the highest loss by a significant margin, suggesting it is less suitable for this task compared to the other models.

4.2.1 Quantitative assessment

The key performance evaluation metrics utilized for the SAR autofocus models include the structural similarity index (SSIM),³³ which measures the structural similarity between two images, assessing how well the luminance, contrast, and structure of the images match. It can be mathematically represented as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \quad (24)$$

where x and y represent the original and output images with local means (μ_x and μ_y), local variances (σ_x^2 and σ_y^2), and local covariance (σ_{xy}) in specific image regions. Constants C_1 and C_2 are introduced to ensure stable divisions and numerical reliability.

Another evaluation metric is the peak signal-to-noise ratio (PSNR), which measures the ratio between the maximum possible power of a signal and the power of corrupting noise that affects the fidelity of its representation.³⁴ The higher the PSNR is, the better the quality of the image is. It can be mathematically represented as

$$\text{PSNR}(x, y) = 10 \cdot \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right), \quad (25)$$

where x refers to the original image, y represents the output image, MAX stands for the maximum pixel value, and MSE denotes the mean squared error, a metric used to measure the difference between x and y in image processing, with lower values signifying better similarity.

In the research, a comparative analysis is conducted of the proposed ResU-Net model against established state-of-the-art models within the domain of SAR autofocus technology. Detailed in Table 2, this comparison involves the performance of each model across different terrains—mountains, sand, and sea—utilizing average SSIM and PSNR as the key metrics for

Table 2 Quantitative assessment results for different terrains.

Model	Mountains		Sand		Sea	
	Average PSNR (dB)	Average SSIM	Average PSNR (dB)	Average SSIM	Average PSNR (dB)	Average SSIM
GAN network	16.41	0.70	14.50	0.65	18.77	0.78
Sparse AE	18.80	0.81	17.63	0.78	21.10	0.83
CAE	20.48	0.84	19.05	0.80	22.70	0.88
U-Net	22.60	0.89	21.11	0.85	26.03	0.96
ResU-Net	24.78	0.93	23.40	0.90	24.81	0.94

The bold values indicate the highest average PSNR and SSIM of deep-learning SAR-focusing models.

evaluation. This comprehensive assessment serves to highlight the capabilities and limitations of each model with the intricate requirements of SAR image processing.

The evaluation metrics detailed in Table 2 present a comprehensive analysis of the performance of deep-learning focusing models across various terrains. The ResU-Net model emerges as a generally superior architecture, yielding the highest average SSIM and PSNR across mountains (0.93 SSIM and 24.78 PSNR dB) and sand (0.90 SSIM and 23.40 PSNR dB), confirming its robustness and adaptability. This performance is attributed to its sophisticated design that enhances the U-Net framework with residual learning. It is particularly effective in capturing the intricate details of both rugged and flat terrains.

In the sea category, however, the U-Net model outperforms the advanced ResU-Net with an exceptional average SSIM of 0.96 and PSNR of 26.03 dB. This indicates that U-Net's structure, while simpler than that of ResU-Net, is highly efficient in handling the homogeneity of sea images, where less complexity allows for a clearer distinction of features and anomalies. ResU-Net's advanced architecture is less beneficial for simple "sea" images due to their uniformity, risking overfitting by learning noise. However, it excels with "mountain" and "sand" images, where its depth and residual connections capture complex textures, enhancing performance. Essentially, its capability to learn detailed patterns becomes a limitation with the minimal features of sea imagery.

The CAE and SAE models both exhibit distinct trade-offs across different terrains. The CAE demonstrates consistent performance, with its encoding–decoding mechanism yielding an average SSIM of 0.84 and PSNR of 20.48 dB in mountainous terrains, indicating a solid capability in image reconstruction. This performance slightly decreases in sandy environments, with SSIM and PSNR values dropping to 0.80 and 19.05 dB, respectively. Although the CAE shows competent encoding and decoding abilities, it falls short in capturing spatial relationships as effectively as U-Net-based models. On the other hand, the SAE, which prioritizes feature sparsity for a more compact representation, achieves moderate average SSIM and PSNR values of 0.81 and 18.80 dB, respectively, in mountainous regions. Although the SAE maintains a comparable level of detail retention across sand and sea terrains, this approach results in some loss of detailed information due to its focus on reducing the number of active neurons, which might lead to a loss of some critical image details.

Finally, the GANs model, which is primarily designed for generating new images, exhibits the least effective performance in SAR image reconstruction for sand terrain, with an SSIM of 0.65 and a PSNR of 14.50 dB. However, in sea terrain, it achieves an SSIM of 0.78 and a PSNR of 18.77 dB. Despite these gains, it remains deficient in other models, highlighting its fundamental design constraints for tasks that value the fidelity of image reconstruction more than the creation of new images.

Overall, the collective analysis underscores the significance of selecting a model adept at handling the specific characteristics of the target terrain. The nuanced differences in performance across mountains, sand, and sea terrains highlight the impact of environmental factors on the models' reconstruction capabilities. This emphasizes the need for a tailored approach when

applying these models to real-world SAR imaging tasks, ensuring that the chosen model aligns with the complexities and challenges of the respective terrain for optimal image processing outcomes.

To represent the performance of various state-of-the-art models in comparison to the ResU-Net model, improvement ratios are utilized. These ratios are based on the overall average SSIM and PSNR metrics. An improvement ratio indicates the relative performance between models; the lower the improvement ratio is, the closer the model's performance is to ResU-Net, and the higher the improvement ratio is, the lower the model's performance is. The improvement ratio can be calculated using the following equation:

$$\text{improvement ratio}(\%) = \left(\frac{\text{Metric}_{\text{ResU-Net}} - \text{Metric}_{\text{comparative model}}}{\text{Metric}_{\text{ResU-Net}}} \right) \times 100. \quad (26)$$

Here $\text{Metric}_{\text{ResU-Net}}$ represents the overall average metric value (either SSIM or PSNR) for the ResU-Net model. $\text{Metric}_{\text{comparative model}}$ represents the overall average metric value (either SSIM or PSNR) for the model being compared to the ResU-Net.

As depicted in Fig. 15, the improvement is measured in terms of overall average PSNR and average SSIM and is presented as a percentage. When compared to the proposed ResU-Net, indicating that the U-Net is the closest in the performance to the ResU-Net, following the U-Net, the convolutional AE, then the sparse AE, and finally GANs. This visualization emphasizes the ResU-Net's advancements compared to its counterparts, showcasing its potential as a state-of-the-art solution for SAR autofocus tasks and confirming its contribution to advancing the field through significant enhancements in image clarity and noise reduction.

To assess the performance of the CNN-based classification model, the confusion matrix is used when testing the model on test data where the true values are known. The evaluation metrics, such as accuracy, sensitivity/recall, and precision, are calculated from the confusion matrix to understand the performance and behavior of the classification model. The confusion matrix is illustrated in Table 3.

The confusion matrix indicates a high number of true positives (TP), low false positives (FP), and false negatives (FN). From these values, the proposed model reaches an overall accuracy (OA) of (97.33%), sensitivity (99.32%), and precision (97.99%), which demonstrate the model's strong performance and robustness in accurately distinguishing these landscape categories. This model is highly effective for landscape image classification tasks.

4.2.2 Visual assessment

Visual assessment is an essential tool for evaluating the quality of SAR autofocus and the fidelity of processed images, highlighting the effectiveness of the proposed methodologies. The

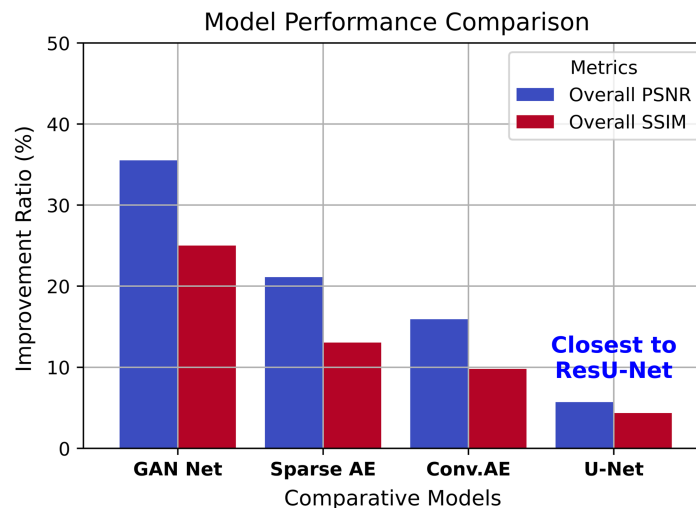


Fig. 15 Comparison of improvement ratios: U-Net closely matches our proposed ResU-Net.

Table 3 Confusion Matrix for CNN-based classification model.

	Predicted: mountains	Predicted: sand	Predicted: sea
Actual: mountains	49 (TP)	1 (FP)	0 (FP)
Actual: sand	3 (FN)	47 (TP)	0 (FP)
Actual: sea	0 (FN)	0 (FN)	50 (TP)

proficiency of the ResU-Net model is particularly evident when it is compared with the ground truth images derived from the traditional RDA and with state-of-the-art models, such as the CAE, SAE, and GANs. For a more detailed and feature-rich comparison, these visual assessments focus on mountain scenes, which inherently contain more intricate features than sea or sand landscapes. Accompanied by the corresponding PSNR and SSIM values, the images generated by the ResU-Net model demonstrate superior capability in feature extraction as shown in Fig. 16. The structures within the ResU-Net images exhibit greater clarity and closely resemble the ground truth data. Through the architectural advancements of ResU-Net, these comparisons not only underscore its superior performance in SAR image processing but also its enhanced capability to preserve and enhance image details, especially in complex terrains like mountain scenes. This refined visual assessment strategy provides a comprehensive understanding of the model’s performance, ensuring a thorough evaluation of its effectiveness in extracting and refining features from challenging landscapes.

To sum up, our comprehensive comparative analysis rigorously evaluated the ResU-Net model against established SAR autofocus models, such as the CAE, SAE, and GANs, utilizing the same datasets to ensure a fair comparison. The results highlight the ResU-Net’s superior performance, which can be attributed to its inherent architectural innovations rather than dataset

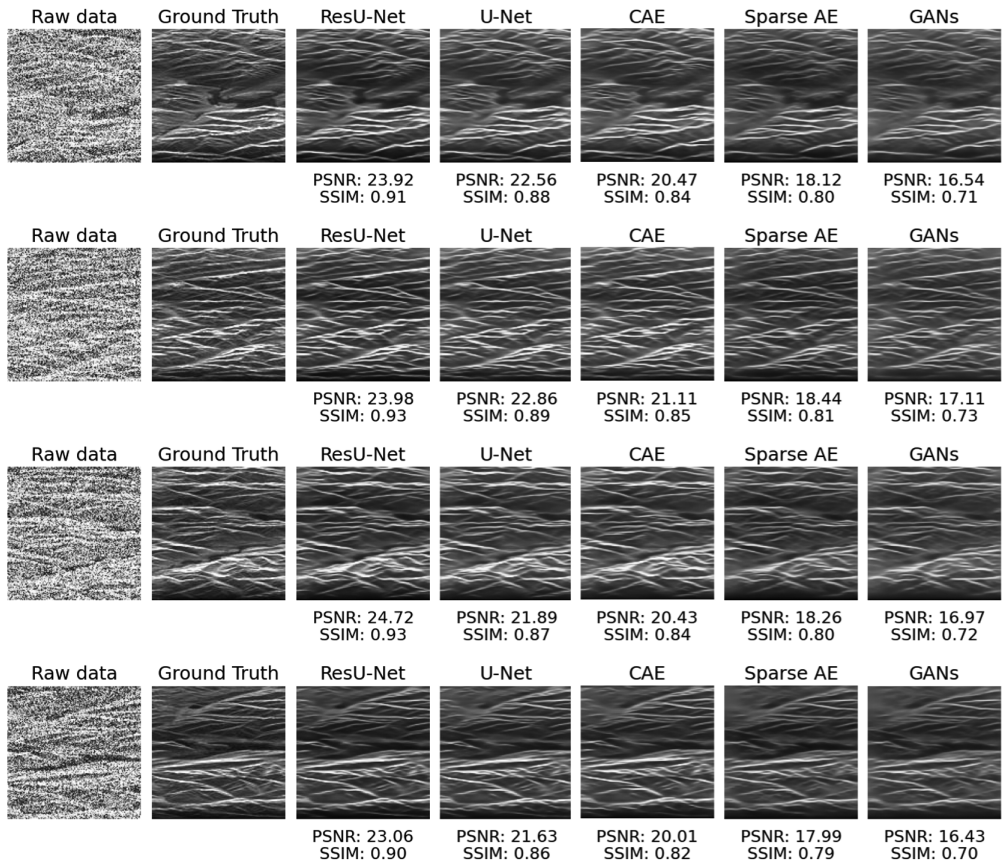


Fig. 16 Comparative visual assessment of SAR image autofocus models.

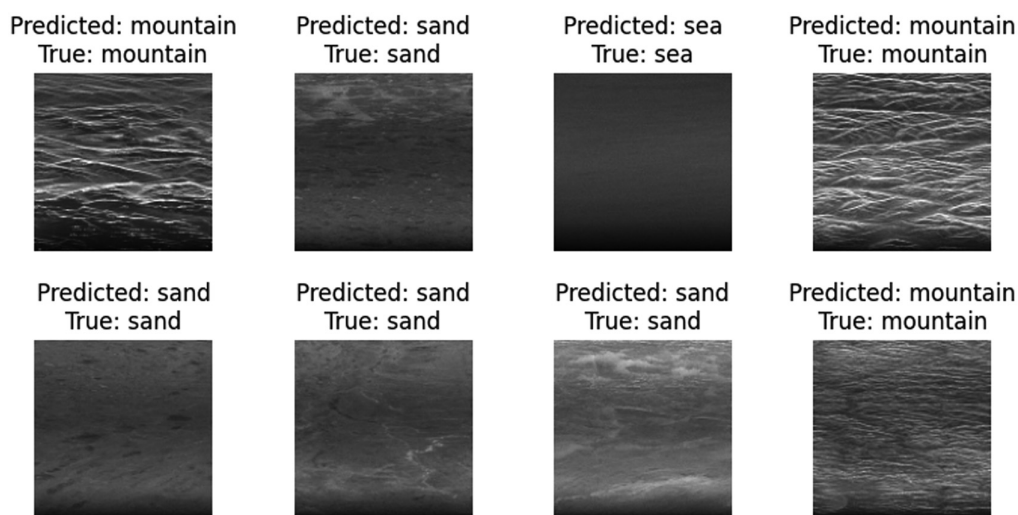
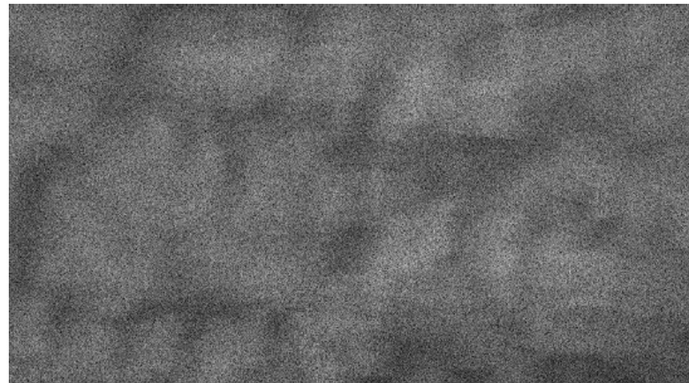


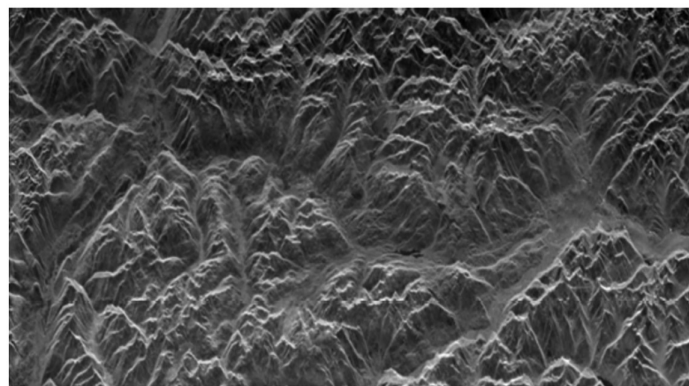
Fig. 17 CNN model's proficiency in landscape classification.

specificity. This superiority demonstrated through both quantitative and qualitative assessments underscores the model's robustness and its enhanced capability for detailed image reconstruction across varied conditions. These insights not only underscore the efficacy of the ResU-Net model but also pave the way for future explorations in the application of deep learning for enhanced SAR imagery, marking a pivotal progression in the fusion of deep learning with remote sensing technologies.

As shown in Fig. 17, the CNN-based classification model is depicted through the set of images, showcasing its proficiency in making predictions. This model distinguishes intricate



(a)



(b)

Fig. 18 Real-case scenario: (a) raw image data and (b) focused SAR image.

terrains, as evident in the visual outputs, effectively categorizing landscapes into mountains, sand, or sea.

The real-case scenario presented in Fig. 18 illustrates the proficiency of the complex 2D autofocus SARNet pipeline in converting raw SAR data into a focused and classified SAR image. This instance, capturing a scene frame that spans a 100 km swath in Canada, identified as the AOI, exemplifies the advanced computational capabilities of the pipeline. The transformation of raw data into a focused image was accomplished within a remarkable time frame of 71.43 ms. This process involved segmenting the raw data, focusing each segment, and then reassembling these focused segments into a single image. This achievement underscores the pipeline’s applicability for near-real-time operations in image visualization and scene classification, demonstrating its significant potential in the field of remote sensing.

4.3 Computational Efficiency

As illustrated in Table 4, the integrated SAR processing pipeline demonstrates a significant enhancement in processing speed, being ~50 times faster than the traditional SAR autofocus (RDA). This marked increase in efficiency is particularly critical for time-sensitive tasks. The enhanced performance can be attributed to the employment of advanced hardware and software technologies, including an NVIDIA RTX A2000 GPU, an Intel® Core™ i9-10900k CPU, and 32 GB of RAM. Furthermore, the integration with the TensorFlow-included Keras framework likely contributes to the efficiency and speed of the processing pipeline. This fusion between high-performance computing resources and optimized software frameworks underpins the substantial improvements in processing speed for SAR data analysis.

To summarize our insights, the proposed pipeline exhibits several strengths that contribute to its outstanding performance in the SAR image processing task.

- *Utilization of ResU-Net architectures incorporating residual blocks.* This enhances feature learning and extraction by incorporating residual blocks into the encoder path of the U-Net architecture. This significantly improves image formation efficiency, aiding in rapid processing and facilitating complex imaging task handling.
- *Robustness and efficiency for CNN-based classification model.* This exhibits robustness and efficiency in scene classification across diverse terrains. This capability underscores the model’s adaptability and reliability in accurately interpreting varied landscape features, enhancing its utility for comprehensive terrain analysis.
- *Significant speed enhancement in processing.* The integrated SAR processing pipeline achieves a processing speed ~50 times faster than traditional SAR autofocus (RDA) methods. This significant enhancement in speed is crucial for time-sensitive tasks, ensuring rapid data analysis and decision-making capabilities.
- *Efficient handling of massive SAR data volumes.* This demonstrates capability to process large datasets swiftly, facilitating near-real-time image viewing and scene classification.
- *Real-SAR data validation.* This ensures reliability and robustness through extensive testing with actual SAR imagery.
- *Comprehensive performance evaluation.* This uses standard metrics to objectively assess the system’s efficiency, reliability, and responsiveness.

Table 4 Computational efficiency of SAR processing.

Feature	Complex 2D-SARNet	Traditional SAR autofocus (RDA)
Processing rate (frames per second)	14 FPS	0.28 FPS (1 frame per 3.56 s)
Relative speed (compared to traditional)	50× faster (approx.)	1× (baseline)
Real-time capability	Yes (highly effective)	No (slower processing)
Suitability for time-sensitive tasks	Highly suitable	Less suitable

The 2D-SARNet framework has shown promising advancements in SAR image processing, albeit with certain challenges related to SAR data characteristics and computational requirements. These challenges are addressed as follows.

- *Limited diversity in training data.* The model's robustness and generalizability for different sensors are affected by the diversity of the training datasets, which is constrained by the availability of raw SAR sensor data. To address this, we emphasize the need to include raw SAR data from various satellites, which is not always available. So we will expand our dataset "ERS satellite raw data"^{19,20} with other raw SAR sensor data in future work.
- *Computational overhead and resource challenges.* The characteristics of the dataset utilized for model training are represented in data volume and image size. High data volumes require significant computational power for processing, which is constrained by the available resources (processors-GPUs-... etc.). To address these challenges, our study optimizes data handling by dividing the scene frame into six segments and downsampling each to 256×256 for efficient processing, as detailed in Sec. 4.1. This approach, which was previously utilized and verified by research from Rittenbach and Walters¹² and Huo et al.,¹¹ efficiently manages resources while maintaining image quality.
- *Dependence on input data quality.* The performance of our 2D-SARNet model is influenced by the quality of SAR sensor data. To tackle issues with inaccurate or incomplete data as shown in Fig. 19, we implement rigorous data cleaning, preprocessing, and validation measures to ensure data integrity. We train our model on high-quality datasets and actively exclude any inaccurate or incomplete data. These measures significantly improve the model's accuracy and robustness, making it reliable for various SAR image processing applications.
- *Sensitivity to noise and errors in data.* Considering that the primary objective of our proposed model is to enhance SAR image formation and autofocus, the selection of an appropriate dataset for model training is crucial. To ensure optimal model performance, it is essential to use a dataset that is as free from noise as possible. To address these challenges, we strategically exclude noisy data, as illustrated in Fig. 20, from our training set. This exclusion is accomplished by initially forming images using the traditional RDA. We then evaluate these images to identify and remove those with undesirable noise levels before they are used in training our model.

Overall, this integrated SAR pipeline improves image quality compared with other models and demonstrates robust image classification capabilities, effectively differentiating between various landscape categories. Therefore, the combination of the ResU-Net and the CNN-based classification model offers a comprehensive and efficient solution for SAR image processing, ensuring both superior image quality and precise image classification.

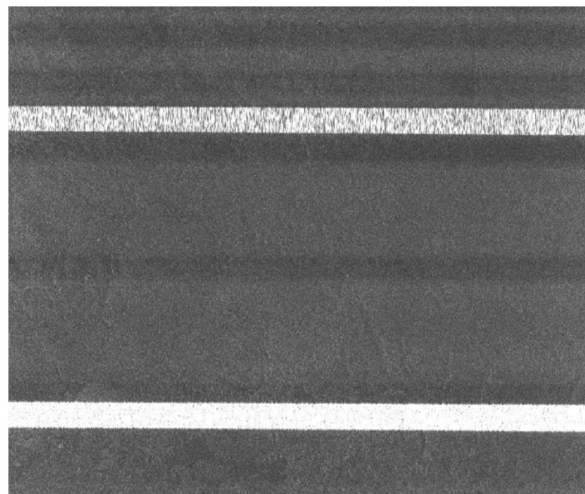


Fig. 19 Example of inaccurate or incomplete input raw data from ERS SAR sensor.

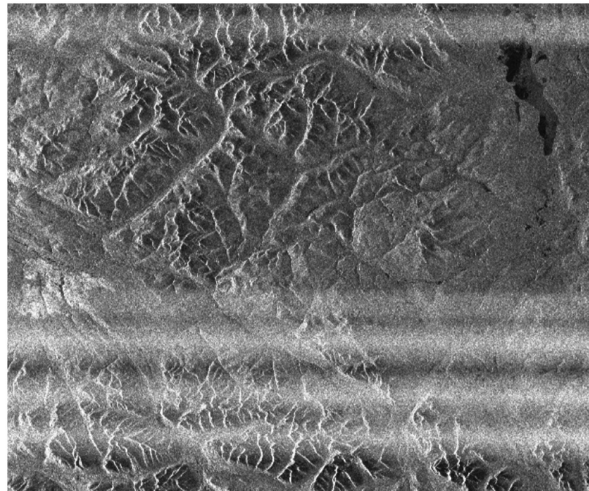


Fig. 20 Example of high noise level in ERS image.

5 Conclusion

This paper presents complex 2D-SARNet, a deep-learning framework adapted for SAR image processing, which combines a ResU-Net for enhanced autofocus and a CNN for terrain-type classification. This framework significantly advances the processing of large SAR data volumes, substantially boosting autofocus accuracy and speed up to 50 times faster than traditional methods like RDA, through comprehensive experiments on real-satellite data. The performance of the SAR pipeline is evaluated using quantitative metrics, such as an average SSIM of 0.92, a PSNR of 24.74 dB for the ResU-Net, and a 97.33% accuracy rate for the CNN model. These measures along with visual evaluations confirm the pipeline's effectiveness in generating SAR-focused images and its reliability in accurately distinguishing between various landscapes, including mountainous, sandy, and marine environments. A comparative study with available state-of-the-art models highlights the proposed method's superior performance. However, the approach faces challenges such as the need for more diverse training datasets, high computational requirements, and sensitivity to data quality and noise, which could affect its scalability and generalization capabilities. By leveraging deep learning alongside SAR signal processing, this integrated method facilitates near-real-time analysis vital for time-sensitive applications, such as disaster response, surveillance, and mapping, showcasing deep learning's potential to overcome traditional SAR processing obstacles.

6 Future Work

This project initially focused on ERS satellite images. In future endeavors, we plan to enhance the robustness and generalizability of our models by incorporating SAR data from various satellite systems, including high-resolution images that permit the use of advanced deep-learning techniques. By broadening our dataset, we aim to improve our SAR image processing capabilities significantly, enabling more precise analysis of complex terrains and environmental features. To overcome the inherent noise and error sensitivities in SAR data, we will employ innovative machine-learning strategies such as GANs, convolutional denoising autoencoders, and reinforcement learning. These efforts are designed to boost our preprocessing effectiveness, achieving better noise reduction and error correction, and ultimately developing a resilient and adaptable SAR image processing framework that delivers high-quality results in a range of real-world applications.

Disclosures

The authors declare no conflicts of interest.

Code and Data Availability

The raw SAR echo datasets utilized in this study were obtained from the ERS satellite and are publicly available. They were accessed via the ASF using ASF's Vertex program and can be obtained from their respective sources. The ERS mission summary can be accessed from [<https://earth.esa.int/eogateway/missions/ers/description>]. SAR raw data from the ERS satellite can be accessed from [<https://search.asf.alaska.edu/#/?dataset=ERS>]. This dataset contains raw SAR echo data and is widely used for various remote sensing applications. The ASF Vertex tool, which was used to download frames of raw SAR data collected by the ERS satellite, can be accessed at [<https://www.asf.alaska.edu>]. It is a graphical search interface for finding and downloading SAR Data. Researchers interested in replicating or validating our study are encouraged to obtain these datasets from the provided sources.

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References

1. H. Cruz et al., "A review of synthetic-aperture radar image formation algorithms and implementations: a computational perspective," *Remote Sens.* **14**(5), 1258 (2022).
2. I. G. Cumming and F. H. Wong, *Digital Processing of Synthetic Aperture Radar Data*, Vol. 1, pp. 108–110, Artech House (2005).
3. A. Moreira et al., "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.* **1**(1), 6–43 (2013).
4. A. Reigber et al., "Very-high-resolution airborne synthetic aperture radar imaging: signal processing and applications," *Proc. IEEE* **101**(3), 759–783 (2012).
5. W. Fan et al., "Modified range-Doppler algorithm for high squint SAR echo processing," *IEEE Geosci. Remote Sens. Lett.* **16**(3), 422–426 (2018).
6. Q. Xiaolan, H. Donghui, and D. Chibiao, "Non-linear chirp scaling algorithm for one-stationary bistatic SAR," in *1st Asian and Pac. Conf. Synthetic Aperture Radar*, IEEE, pp. 111–114 (2007).
7. M. M. Ashry, A. S. Mashaly, and B. I. Sheta, "Improved SAR range Doppler algorithm based on the stretch processing architecture," in *Int. Telecommun. Conf. (ITC-Egypt)*, IEEE, pp. 1–6 (2022).
8. M. M. Taye, "Understanding of machine learning with deep learning: architectures, workflow, applications and future directions," *Computers* **12**(5), 91 (2023).
9. M. I. Mohamed, K. M. Badran, and A. E. Hussien, "Anomaly detection for agile satellite attitude control system using hybrid deep-learning technique," *J. Aerosp. Inf. Syst.* **20**(12), 890–904 (2023).
10. M. AboElenean et al., "Land cover analysis of polSAR images using probabilistic voting ensemble and integrated support vector machine," *J. Appl. Remote Sens.* **17**(4), 044505 (2023).
11. J. Huo et al., "Novel SAR autofocus method based on deep learning with GAN network," in *CIE Int. Conf. Radar (Radar)*, IEEE, pp. 800–803 (2021).
12. A. Rittenbach and J. P. Walters, "RDANet: a deep learning based approach for synthetic aperture radar image formation," arXiv:2001.08202 (2020).
13. W. Pu, "SAE-Net: a deep neural network for SAR autofocus," *IEEE Trans. Geosci. Remote Sens.* **60**, 1–14 (2022).
14. H. Bi et al., "From theory to application: real-time sparse SAR imaging," *IEEE Trans. Geosci. Remote Sens.* **58**(4), 2928–2936 (2019).
15. M. Sakr et al., "Enhanced range-Doppler algorithm for SAR image formation," in *J. Phys.: Conf. Ser.*, IOP Publishing, Vol. 2616, p. 012033 (2023).
16. S. Baraha and A. K. Sahoo, "Synthetic aperture radar image and its despeckling using variational methods: a review of recent trends," *Signal Process.* **212**, 109156 (2023).
17. O. Ronneberger, P. Fischer, and T. Brox, "U-Net: convolutional networks for biomedical image segmentation," *Lect. Notes Comput. Sci.* **9351**, 234–241 (2015).
18. S. Jadon, "A survey of loss functions for semantic segmentation," in *IEEE Conf. Comput. Intell. in Bioinform. and Comput. Biol. (CIBCB)*, IEEE, pp. 1–7 (2020).

19. European Space Agency (ESA), "ERS mission summary," <https://earth.esa.int/web/guest/missions/esa-eomissions/ers/mission-summary> (Open Accessed).
20. A. S. Facility, "Alaska satellite facility," <https://www.asf.alaska.edu> (Open Accessed).
21. Alaska Satellite Facility, "ASF vertex tool," <https://search.asf.alaska.edu/> (Open Accessed).
22. L. A. Gorham and L. J. Moore, "SAR image formation toolbox for MATLAB," *Proc. SPIE* **7699**, 769906 (2010).
23. T. Yang et al., "ISAR imaging enhancement: exploiting deep convolutional neural network for signal reconstruction," *Int. J. Remote Sens.* **41**(24), 9447–9468 (2020).
24. X. Miao et al., "Automatic recognition of highway tunnel defects based on an improved U-Net model," *IEEE Sens. J.* **19**(23), 11413–11423 (2019).
25. A. K. Dubey and V. Jain, "Comparative study of convolution neural network's relu and leaky-relu activation functions," in *Appl. of Comput., Autom. and Wirel. Syst. in Electr. Eng.: Proc. MARC*, Springer, pp. 873–880 (2019).
26. S. Sharma, S. Sharma, and A. Athaiya, "Activation functions in neural networks," *Towards Data Sci.* **6**(12), 310–316 (2017).
27. W. Song et al., "An improved U-Net convolutional networks for seabed mineral image segmentation," *IEEE Access* **7**, 82744–82752 (2019).
28. H. Shahid et al., "A deep learning approach for the photoacoustic tomography recovery from undersampled measurements," *Front. Neurosci.* **15**, 598693 (2021).
29. H. Shahid et al., "Batch renormalization accumulated residual u-network for artifacts removal in photoacoustic imaging," in *IEEE Int. Ultrason. Symp. (IUS)*, IEEE, pp. 1–4 (2021).
30. J. Ding et al., "Convolutional neural network with data augmentation for SAR target recognition," *IEEE Geosci. Remote Sens. Lett.* **13**(3), 364–368 (2016).
31. Y. Dong et al., "Impact analysis of incident angle factor on high-resolution SAR image ship classification based on deep learning," in *IGARSS 2019-2019 IEEE Int. Geosci. and Remote Sens. Symp.*, IEEE, pp. 1358–1361 (2019).
32. D. P. Kingma and J. Ba, "Adam: a method for stochastic optimization," arXiv:1412.6980 (2014).
33. Z. Wang et al., "Image quality assessment: from error visibility to structural similarity," *IEEE Trans. Image Process.* **13**(4), 600–612 (2004).
34. P. Wang, H. Zhang, and V. M. Patel, "SAR image despeckling using a convolutional neural network," *IEEE Signal Process. Lett.* **24**(12), 1763–1767 (2017).

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